GenoPhylax

A Unified Cyber-Biosecurity Framework Across the Genomic Data Lifecycle

GenoPhylax studies security failures across the full genomic data lifecycle — from sequencing and basecalling, through alignment and clinical interpretation, to storage, sharing, privacy amplification, and containment — as a single connected threat surface rather than isolated problems.

Six experimental modules feed into a lifecycle-integration layer:

- **A1–A3:** generation, processing, and clinical-integrity risks
- **B1–B3:** storage, privacy, amplification, containment, and authentication
- **Module C:** lifecycle integration and the **GenoPhylax Risk Index (GRI)**

Authoritative Module C implementation:
`experiments/C_lifecycle_integration/authoritative_rebuild/`

Final authoritative integration commit: `d1a7de5` — `Rebuild Module C from authoritative lifecycle evidence`

1. Core Research Question and Contribution

Core research question: Can heterogeneous security failures across genomic generation, processing, storage, sharing, and reuse be represented as evidence-backed lifecycle chains rather than as isolated vulnerabilities?

GenoPhylax evaluates this through two principal integrated cases:

1. a **clinical-integrity chain**, where a targeted nanopore signal perturbation propagates through allele evidence and downstream variant calling; and
2. a **privacy-amplification chain**, where genomic disclosure produces persistent re-identification risk, familial exposure, kinship amplification, and topology-dependent containment difficulty.

The framework integrates only semantically justified evidence into the GRI while keeping validation controls, supporting evidence, standalone security findings, and mitigation evidence separate.

2. Canonical Genomic Lifecycle (S1–S10)

Stage	Name	Description

S1	Generation	Raw sequencing-data generation
S2	Basecalling	Raw signal → nucleotide sequence
S3	Alignment	Reads mapped to a reference genome
S4	Variant Calling	Allele/variant evidence extraction
S5	Clinical Interpretation	Variants → clinical meaning
S6	Storage	Persistence in repositories or vaults
S7	Sharing / Release	Disclosure of genomic information
S8	Reuse / Downstream Propagation	Secondary analysis, kinship inference, or amplification
S9	Incident / Forensic Analysis	Detection and investigation of security events
S10	Containment / Revocation	Controls limiting further propagation

3. Module A1 — BWA Memory-Safety Audit

Target: BWA 0.7.19-r1273
Method: BWA compiled with AddressSanitizer and fuzzed using AFL++ with malformed genomic input.

Campaign

Executions: 963,339

Runtime: 23,120 s (~6.42 h)

Bitmap coverage: 15.44%

Crash artifacts: 521

Findings

Two distinct source-level out-of-bounds memory-read findings were identified:

- `ksw_extend2` — `ksw.c:428`
- `mem_aln2sam` — `bwamem.c:906`

Scope

These findings do **not** establish:

- remote code execution;
- clinical compromise;
- patient harm;
- a demonstrated exploit chain.

No experimental path connects A1 directly to either the F11 clinical case or the privacy model.

GRI role: `STANDALONE_NON_GRI`

A1 is preserved in the evidence architecture as an independent genomic software-security result but is not scored into Base GRI.

4. Module A2 — Adversarial Nanopore Signal Experiments

Setup: HG002 / GM24385, Oxford Nanopore R10.4.1 E8.2 400 bps, Dorado basecaller using:

`dna_r10.4.1_e8.2_400bps_hac@v6.0.0`

Bonito was evaluated earlier but was not retained in the final practical pipeline.

Clean baseline

Runtime: 27.62 s

Mean Q-score: 35.03

Quickcheck: PASS

Gaussian perturbation dose-response

Mean sequence identity decreased as perturbation strength increased:

GN01: 98.7903%

GN05: 96.9033%

GN10: 93.8863%

The experiments showed measurable dose-dependent basecalling degradation while substantial mapping could still be preserved.

Cross-locus W0 specificity control

Independent loci:

L2: chr1:20061156 A>T

L3: chr4:40028853 A>G

Result:

L2: 10/10 parents retained ALT

L3: 10/10 parents retained ALT

Combined target-state changes: 0/20

Findings

A2 demonstrates that:

- controlled signal perturbation measurably alters basecalled sequence;
- degradation scales with perturbation strength;
- reads can remain sufficiently intact to continue downstream;
- local perturbation effects can be distinguished from global read destruction;
- cross-locus controls are necessary before claiming targeted biological consequence.

The targeted perturbation experiments established the experimental basis for the subsequent F11 clinical-consequence case.

GRI role: `SUPPORTING_NON_GRI`

5. Module A3 — Clinical Consequence Mapping: F11 Case

Research question

Can a controlled signal perturbation propagate far enough through the genomic pipeline to alter clinically relevant variant evidence?

Target

Chromosome: chr4

Position: 186274193

Variant: G>T

Gene: F11

ClinVar Variation ID: 11891

Sample: HG002

Clean condition

Parent reads: 15

Dorado records: 16

Mapped records: 16

Primary mapped: 16

Among evaluable parent reads:

ALT retention: 14/14

Targeted PM2 perturbation

Clean ALT-supporting parents: 14/14

PM2 ALT-supporting parents: 0/14

Downstream VCF result: absent / NO_CALL

The targeted perturbation therefore produced complete loss of evaluable ALT support in the tested case and propagated to downstream loss of the F11 variant call.

Matched off-target sham

ALT-supporting parents: 14/14

Genotype: 0/1

QUAL: ~222.202

The F11 variant remained present under the off-target sham.

This supports localization of the effect rather than generic pipeline degradation caused by applying the perturbation procedure.

Demonstrated lifecycle chain

Targeted nanopore signal perturbation

↓

Loss of target ALT evidence

↓

Loss of downstream variant call

↓

Changed clinical variant representation

Lifecycle handoffs:

HF11_01: S2 → S4

HF11_02: S4 → S5

Clean and off-target conditions are retained as validation controls rather than independent risk contributions.

Scope

The experiment demonstrates **possibility and controlled propagation**, not prevalence.

It consists of:

one HG002 individual

one pathogenic F11 locus

one primary basecaller configuration

one controlled perturbation design

It does not establish generalization to other loci, samples, basecallers, diseases, or patient outcomes.

6. Module B1 — Kinship Graph Amplification

Graph

Nodes: 4,144

Edges: 157,147

Registered: 1,657

The tested DTC registration level was approximately 40%.

Amplification

Raw uncalibrated amplification: 2110×

External reference target: 492.86×

Best cousin probability: 0.20

Calibrated mean amplification: 464.4×

Absolute difference: 28.46×

Relative difference: ~5.77%

The `2110×` value is the raw uncalibrated primary simulation result and is **not** the final calibrated amplification value.

The calibrated model produced **464.4×** mean amplification against the **492.86× external reference target used by the model**, a relative difference of approximately **5.77%**.

D6 score

D6 score: 8.98 / 10

Normalized D6: 0.898

Finding

B1 models the relational nature of genomic privacy: disclosure of one individual's genomic information can create exposure for biologically related individuals through kinship structure. The magnitude of amplification remains dependent on graph construction and calibration assumptions.

7. Module B2 — Genomic Vault and Lifetime Privacy Risk

Objective

B2 studies persistent privacy risk following genomic disclosure.

The model evaluates:

- privacy unrecoverability;
- cumulative disclosure;
- re-identification;
- familial cascade;
- disease inference;

- long-horizon persistence.

Canonical clinician disclosure path

Pathogenic Variants 0.2299

+ Polygenic Risk Scores 0.1346

+ Pharmacogenomic Markers 0.0994

+ Ancestry Composition 0.0864

+ Trait Predictions 0.0811

+ Aggregate Statistics 0.0801

Canonical value used in Module C:

Pathogenic-variant unrecoverability = 0.2299

Other disclosure orderings and scenarios exist in B2 and are not conflated with this canonical clinician path.

Lifetime milestones

Risk	25%	50%	75%	100%
Re-identification	Year 8	Year 11	Year 12	Year 13
Familial cascade	Year 8	Year 11	Year 13	Year 15
Disease inference	Year 14	Year 22	Year 33	Not reached

D5 persistence trajectory

Year 0: 0.75

Year 10: 3.70

Year 20: 8.17

Year 30: 9.06

Year 50: 9.33

Year 80: 9.33

The authoritative GRI normalization uses:

D5_year_50 = 9.33 / 10

Finding

The B2 model shows that genomic privacy loss can be cumulative, persistent, difficult to reverse, and relevant to relatives in addition to the original data subject. These values are model outputs and are not longitudinal population measurements.

8. Module B3 — Containment and Authentication

Graph

Nodes: 3,303

Registered: 1,321

B3 uses a different graph realization from B1; the two graph populations should not be interchanged.

Primary containment

Baseline: 786 exposed nodes

k	Exposed	Reduction	Cost
5	560	28.8%	0.38%
10	6	99.2%	0.76%
15	6	99.2%	1.14%
20	6	99.2%	1.51%

30	6	99.2%	—
40	6	99.2%	—
50	6	99.2%	—

Best exact containment reduction: 99.24%

Attack-origin dependence

Origin	Baseline exposed	Reduction at k=10	Minimum tested k for ≥99% reduction
Peripheral	780	99.2%	5
Mid-degree	943	20.5%	20
Hub	1,077	3.0%	20

Finding

Containment effectiveness depends strongly on attack origin.

The primary scenario reaches approximately 99% containment at `k=10`, but this result does not generalize uniformly across graph positions. Mid-degree and hub-origin attacks require substantially larger tested containment budgets.

The `k=20` result should be interpreted as:

> the minimum tested value reaching at least 99% reduction for the mid-degree and hub scenarios in the evaluated B3 experiment.

It is not claimed as a universal graph-theoretic threshold.

M5 residual containment candidates

At fixed `k=10`:

Peripheral: 1 - 0.992 = 0.008

Mid-degree: 1 - 0.205 = 0.795

Hub: 1 - 0.030 = 0.970

Therefore: M5 scenario range = 0.008 ... 0.970

These are scenario-specific modifier candidates and are **not** multiplied into Base GRI.

Authentication

Final credential-stuffing classifier:

Threshold: 3.6

Precision: 0.9070

Recall: 0.7800

F1: 0.8387

Accuracy: 0.9400

TP: 39

FP: 4

TN: 196

FN: 11

Semantic boundary

Authentication F1 measures the classifier's ability to distinguish legitimate and malicious access in the internal benchmark. It does not measure whether a genomic privacy incident or manipulation itself would be detected.

Therefore: Privacy M3 Detectability = UNMEASURED

B3 is retained as: MITIGATION_CONTEXT and does not reduce Base GRI.

9. Module C — Authoritative Lifecycle Integration

The current authoritative Module C implementation is:
`experiments/C_lifecycle_integration/authoritative_rebuild/`

Earlier Module C numerical outputs were superseded after the final Module B audit and are not used as authoritative input to the rebuild.

Final evidence architecture

C2 evidence records: 16

C3 handoff/control rows: 9

Base-GRI handoffs: 5

A1: STANDALONE_NON_GRI

A2: SUPPORTING_NON_GRI

B3: MITIGATION_CONTEXT

10. C0 — Authoritative Source Lock

C0 establishes the input boundary for Module C. Locked upstream sources: 20. Sources include authoritative A1, A2, A3, B1, B2, and B3 evidence plus the validated C1 lifecycle architecture.

Each source is recorded with:

- module;
- role;
- source path;
- purpose;
- file size;
- SHA-256;
- Git HEAD at source lock.

11. C2 — Authoritative Evidence Registry

The C2 registry contains:

Evidence records: 16

GRI-eligible rows: 9

F11 chain: 4

Privacy chain: 8

Standalone A1: 2

Supporting A2: 2

Evidence roles:

STANDALONE_NON_GRI Used for A1.

SUPPORTING_NON_GRI Used for A2.

GRI_ELIGIBLE Used only when evidence has a justified mapping to a defined GRI dimension.

MITIGATION_CONTEXT Used for B3 containment and authentication.

C2 safeguards

A1 direct GRI use: NO

A2 direct Base GRI use: NO

B3 authentication as privacy detectability: NO

Evidence tiers as numerical GRI weights: NO

12. C3 — Evidence-Backed Handoffs

Total handoff/control rows: 9

Breakdown:

Base-GRI-eligible handoffs: 5

F11 risk handoffs: 2

Privacy risk handoffs: 3

Validation controls: 2

Mitigation-context rows: 2

F11 risk handoffs

HF11_01: S2 → S4

HF11_02: S4 → S5

F11 controls

CF11_01: clean control

CF11_02: off-target sham

Controls are retained for validation and do not become independent risk contributions.

Privacy handoffs

HPRIV_01: S6 → S7

HPRIV_02: S7 → S8

HPRIV_03: S7 → S8

These capture disclosure, persistence/familial propagation, and kinship amplification.

Mitigation handoffs

HPRIV_04: S8 → S10

HPRIV_05: S6 → S10

These capture containment and authentication context and are not Base GRI inputs.

13. GenoPhylax Risk Dimensions

Seven dimensions are defined.

Dimension	Name	Meaning
D1	Identity Risk	Re-identification following genomic disclosure
D2	Disease Inference / Clinical Interpretation Risk	Clinically relevant consequence of genomic manipulation or disclosure
D3	Familial Cascade Risk	Risk propagated to biological relatives
D4	Accessibility Risk	Inappropriate or unauthorized genomic access
D5	Persistence Risk	Difficulty reversing genomic exposure over time
D6	Amplification Risk	Expansion through sharing, kinship, or downstream reuse

D7	Pipeline Exposure Risk	Propagation of upstream integrity failure through downstream genomic processing

Not every dimension applies to every case. Non-applicable or unmeasured dimensions are excluded rather than assigned zero.

14. GenoPhylax Risk Index

For case (c), the Base GRI is the arithmetic mean of applicable, evidence-supported dimension scores:

GRI(c) = sum(applicable dimension scores) / number of applicable dimensions

Framework rules:

Missing ≠ zero

N/A ≠ zero

Evidence tier ≠ numerical multiplier

Mitigation modifier ≠ automatic Base GRI multiplier

Sensitivity result ≠ confidence interval

15. F11 Base GRI

Applicable dimensions:

D2 — Clinical Interpretation Risk

D7 — Pipeline Exposure Risk

D2

Observed chain:

Clean ALT evidence: 14/14

Target PM2 ALT: 0/14

Downstream F11 call: lost

Therefore: D2 = 1.0000

D7

The demonstrated perturbation traversed: S2 → S4 → S5. Therefore: D7 = 1.0000

Base GRI

F11 Base GRI

= mean(D2, D7)

= mean(1.0, 1.0)

= 1.0000000000

Interpretation: This is maximal severity across the two dimensions applicable to this controlled experimental case. It does not represent a 100% probability of patient harm.

16. Privacy Base GRI

Applicable dimensions: D1, D3, D5, D6

D1 — Identity Risk

100% modelled re-identification by year 13. D1 = 1.0000

D3 — Familial Cascade Risk

100% modelled familial exposure by year 15. D3 = 1.0000

D5 — Persistence Risk

Irreversibility component: 1 - 0.2299 = 0.7701

Long-horizon persistence component: 9.33 / 10 = 0.9330

Combined: D5 = mean(0.7701, 0.9330) = 0.85155

The two correlated B2 persistence outputs are combined within one dimension rather than being counted as separate GRI dimensions.

D6 — Amplification Risk

B1 D6 score = 8.98 / 10. D6 = 0.8980

Privacy Base GRI

Privacy Base GRI

= mean(1.0000, 1.0000, 0.85155, 0.8980)

= 0.9373875000

Canonical value: Privacy Base GRI = 0.9373875000

Interpretation: This is an integrated framework-level severity score for the tested/modelled privacy chain. It does not mean that 93.74% of individuals or records will be compromised.

17. Sensitivity and Provenance Weighting

Equal dimension weighting gives:

B2-derived dimensions: D1, D3, D5 = 75%

B1-derived dimension: D6 = 25%

A provenance-balanced sensitivity analysis therefore combines the three B2-derived dimensions into one B2 composite.

B2 composite

mean(D1, D3, D5) = mean(1.0000, 1.0000, 0.85155) ≈ 0.9505167

50/50 provenance weighting

mean(B2 composite, B1 D6) = mean(0.9505167, 0.8980) = 0.9242583333

Canonical sensitivity result: 0.9242583333

The integrated privacy risk remains high after B1 and B2 are given equal total provenance weight.

Additional leave-one-dimension-out analyses are stored in:
`authoritative_rebuild/c4_gri/C4_sensitivity_analysis.tsv`

Sensitivity analysis is a robustness check and is not a confidence interval.

18. Supported Claims

The current authoritative evidence supports the following statements:

- BWA exposed two source-level out-of-bounds memory-read findings under the tested AFL++/ASAN campaign.
- Controlled nanopore signal perturbation produced measurable, dose-dependent basecalling degradation.
- Cross-locus W0 testing produced 0/20 target-state changes under the tested conditions.
- In the controlled HG002 F11 case, targeted PM2 suppressed evaluable ALT support from 14/14 to 0/14.
- The targeted F11 perturbation propagated into downstream loss of the variant call.
- The matched off-target sham preserved ALT evidence and the heterozygous call.
- B1 produced calibrated mean kinship amplification of 464.4× against the 492.86× external reference target used by the model.

- B2 produced pathogenic-variant unrecoverability of 0.2299.
- B2 reached 100% modelled re-identification by year 13 and 100% modelled familial exposure by year 15.
- B3 primary containment achieved 99.24% best reduction.
- B3 containment effectiveness depends strongly on attack origin.
- Mid-degree and hub-origin scenarios required `k=20` to reach ≥99% reduction in the tested parameter sweep.
- B3 authentication produced F1 0.8387 at threshold 3.6.
- F11 Base GRI = 1.0000000000.
- Privacy Base GRI = 0.9373875000.
- Privacy GRI under 50/50 B1-B2 provenance balancing = 0.9242583333.

19. Claims to Avoid

The project should **not** be described as showing that:

- the BWA findings necessarily enable code execution;
- all BWA installations are exploitable;
- all nanopore perturbations cause clinically meaningful errors;
- the F11 result generalizes to all genes, variants, samples, or basecallers;
- F11 Base GRI = 1.0 means a 100% probability of clinical harm;
- a patient's treatment or outcome was altered;
- Privacy Base GRI = 0.9374 means 93.74% of people will be re-identified;
- B1 or B2 outputs are direct population-wide empirical measurements;
- authentication F1 measures privacy-event detectability;
- k=20 is a universal containment requirement for all kinship networks;
- M5 values are universal constants;
- sensitivity analysis constitutes a statistical confidence interval;
- evidence tiers represent calibrated numerical confidence weights.

20. Key Limitations

Experimental scale

The F11 case uses:

one individual

one pathogenic locus

one primary basecaller configuration

Broader replication is required before generalizing the result.

No patient outcome

The project observes:

allele-evidence loss

variant-call loss

changed computational clinical representation

It does not observe actual patient harm.

A2 perturbation scope

A2 uses controlled perturbations and is not a fully optimized adversarial attack generator.

A1 exploitability

The A1 findings are memory-safety findings and do not establish remote exploitation.

Model dependence

B1, B2, and B3 contain synthetic or modelled components.

Results depend on assumptions including:

- graph construction;
- registration rate;
- calibration;
- disclosure model;
- temporal model;
- containment budget;
- attack origin.

GRI validation

The GRI is a framework-level score and has not been externally validated as a clinical or population risk instrument.

Evidence tiers

Evidence tiers communicate evidence class and provenance, not calibrated statistical probabilities.

21. Key Repository Paths

A1 experiments/A1_memory_audit/provenance/

A2 experiments/A2_adversarial_dna/basecalling/findings/master_A2_GPU/

A3 experiments/A3_clinical_consequence_mapping/findings/

Final Module B authority

experiments/B_final_audit/B1_authoritative_final.tsv

experiments/B_final_audit/B2_authoritative_final.tsv

experiments/B_final_audit/B3_authoritative_final.tsv

experiments/B_final_audit/B_authoritative_validation.txt

Authoritative Module C

experiments/C_lifecycle_integration/authoritative_rebuild/

Important Module C files:

inputs/C0_authoritative_sources.tsv

c2_evidence/C2_authoritative_evidence_registry.tsv

c3_handoffs/C3_authoritative_handoffs.tsv

c4_gri/C4_methodology.md

c4_gri/C4_base_gri.tsv

c4_gri/C4_normalized_dimensions.tsv

c4_gri/C4_modifier_context.tsv

c4_gri/C4_sensitivity_analysis.tsv

c5_findings/C5_integrated_findings.md

publication/figures/

publication/tables/

publication/claims/C8_claim_evidence_map.md

22. Canonical Numerical Reference for Paper Drafting

A1

BWA version: 0.7.19-r1273

Executions: 963,339

Runtime: 23,120 s (~6.42 h)

Bitmap coverage: 15.44%

Crash artifacts: 521

Findings:

1. ksw_extend2 — ksw.c:428

2. mem_aln2sam — bwamem.c:906

A2

Sample: HG002

Basecaller: Dorado

GN01 identity: 98.7903%

GN05 identity: 96.9033%

GN10 identity: 93.8863%

Cross-locus W0:

0/20 target-state changes

A3 F11

Variant: chr4:186274193 G>T

Gene: F11

ClinVar Variation ID: 11891

Clean ALT: 14/14

Target PM2 ALT: 0/14

Target call: absent / NO_CALL

Off-target sham:

ALT: 14/14

GT: 0/1

QUAL: ~222.202

B1

Graph nodes: 4,144

Graph edges: 157,147

Registered nodes: 1,657

Raw amplification: 2110×

External reference target: 492.86×

Best cousin probability: 0.20

Calibrated mean: 464.4×

Relative difference: ~5.77%

D6: 8.98/10

Normalized D6: 0.898

B2

Pathogenic-variant unrecoverability: 0.2299

Re-identification 100%: year 13

Familial exposure 100%: year 15

Disease inference 100%: not reached

D5 year 50: 9.33/10

B3

Graph nodes: 3,303

Registered nodes: 1,321

Baseline exposed: 786

k=5:

exposed = 560

reduction = 28.8%

k=10:

exposed = 6

reduction = 99.2%

Best exact reduction = 99.24%

Authentication:

threshold = 3.6

precision = 0.9070

recall = 0.7800

F1 = 0.8387

accuracy = 0.9400

Peripheral:

baseline = 780

k10 reduction = 99.2%

min k99 = 5

Mid-degree:

baseline = 943

k10 reduction = 20.5%

min k99 = 20

Hub:

baseline = 1077

k10 reduction = 3.0%

min k99 = 20

M5:

peripheral = 0.008

mid = 0.795

hub = 0.970

Final GRI

F11:

D2 = 1.0000000000

D7 = 1.0000000000

Base GRI = 1.0000000000

Privacy:

D1 = 1.0000000000

D3 = 1.0000000000

D5 = 0.8515500000

D6 = 0.8980000000

Base GRI = 0.9373875000

Privacy 50/50 B1-B2 provenance sensitivity:

0.9242583333